1. 设 $z(x,y)=\int_0^1 f(t)\,|xy-t|\,\mathrm{d}t$， f 连续， $0\le x,y\le 1$，求 z_{xx}， z_{xy}.

解 （去绝对值，化积分为变限积分）

$$z(x,y)=\int_0^{xy} f(t)(xy-t)\mathrm{d}t+\int_{xy}^1 f(t)(t-xy)\mathrm{d}t$$

$$=xy\int_0^{xy} f(t)\mathrm{d}t-\int_0^{xy} tf(t)\mathrm{d}t+\int_{xy}^1 tf(t)\mathrm{d}t-xy\int_{xy}^1 f(t)\mathrm{d}t\ ,$$

记 $u=xy$，则 $z=u\int_0^u f(t)\mathrm{d}t-\int_0^u tf(t)\mathrm{d}t+\int_u^1 tf(t)\mathrm{d}t-u\int_u^1 f(t)\mathrm{d}t$，

$$\frac{dz}{du}=\int_0^u f(t)\mathrm{d}t+\int_1^u f(t)\mathrm{d}t\ .$$

所以 $z_x=\dfrac{\mathrm{d}z}{\mathrm{d}u}\cdot\dfrac{\partial u}{\partial x}=y\cdot\left(\int_0^u f(t)\mathrm{d}t+\int_1^u f(t)\mathrm{d}t\right)$，

$$z_{xx}=\frac{\partial}{\partial x}\left\{y\cdot\left(\int_0^u f(t)\mathrm{d}t+\int_1^u f(t)\mathrm{d}t\right)\right\}=2y^2 f(xy)$$

$$z_{xy}=\frac{\partial}{\partial y}\left\{y\cdot\left(\int_0^u f(t)\mathrm{d}t+\int_1^u f(t)\mathrm{d}t\right)\right\}$$

$$=\int_0^{xy} f(t)\mathrm{d}t+\int_1^{xy} f(t)\mathrm{d}t+2xyf(xy)\ .$$

注 引入中间变量 u 让计算简明.

2. 设 $u(x,y)$ 具有二阶连续偏导数且满足方程 $u_{xx}=u_{yy}$ 及 $u(x,2x)=x$， $u_x(x,2x)=x^2$，求 $u_{xx}(x,2x),u_{xy}(x,2x)$.

解 $u(x,y)$ 具有二阶连续偏导数，则 $u_{xy}=u_{yx}$.

对 $u(x,2x)=x$ 两边关于 x 求导有 $u_x(x,2x)\cdot 1+u_y(x,2x)\cdot 2=1$，

结合 $u_x(x,2x)=x^2$ 求得 $u_y(x,2x)=\dfrac{1}{2}(1-x^2)$.

再对 $u_x(x,2x)=x^2$， $u_y(x,2x)=\dfrac{1}{2}(1-x^2)$ 两边关于 x 求导有

$$\begin{cases}u_{xx}(x,2x)\cdot 1+u_{xy}(x,2x)\cdot 2=2x\\ u_{yx}(x,2x)\cdot 1+u_{yy}(x,2x)\cdot 2=-x\end{cases}$$

利用 $u_{xx}=u_{yy}$， $u_{xy}=u_{yx}$ 解得 $u_{xx}(x,2x)=-\dfrac{4x}{3},\ u_{xy}(x,2x)=\dfrac{5x}{3}$.

注 偏导记号理解，混合偏导与次序无关.

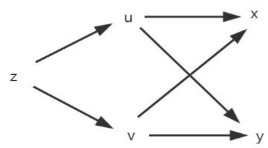
3. 设 $u=xy,v=\dfrac{x}{y}$，以 u,v 为新自变量，变换方程 $x^2z_{xx}-y^2z_{yy}=0$.

解 注意$u=xy, v=\dfrac{x}{y}$可确定$x=x(u,v)=\sqrt{uv}, y=y(u,v)=\sqrt{u}/\sqrt{v}$.

因此方程$x^2 z_{xx}-y^2 z_{yy}=0$中未知函数可理解为

$$z=z(x,y)=z(x(u,v),y(u,v)).$$

这时将z_{xx}, z_{yy} 如下计算（变量关系如图）：



$$z_x=\frac{\partial z}{\partial u}\cdot u_x+\frac{\partial z}{\partial v}\cdot v_x=y\frac{\partial z}{\partial u}+y^{-1}\frac{\partial z}{\partial v},$$

$$z_{xx}=\frac{\partial}{\partial x}(y\frac{\partial z}{\partial u}+y^{-1}\frac{\partial z}{\partial v})=y(y\frac{\partial^2 z}{\partial u^2}+y^{-1}\frac{\partial^2 z}{\partial u\partial v})+y^{-1}(y\frac{\partial^2 z}{\partial v\partial u}+y^{-1}\frac{\partial^2 z}{\partial v^2})$$

$$=y^2\frac{\partial^2 z}{\partial^2 u}+2\frac{\partial^2 z}{\partial u\partial v}+y^{-2}\frac{\partial^2 z}{\partial^2 v};$$

$$z_y=\frac{\partial z}{\partial u}\cdot u_y+\frac{\partial z}{\partial v}\cdot v_y=x\frac{\partial z}{\partial u}-xy^{-2}\frac{\partial z}{\partial v},$$

$$z_{yy}=x^2\frac{\partial^2 z}{\partial^2 u}-2x^2y^{-2}\frac{\partial^2 z}{\partial u\partial v}+x^2y^{-4}\frac{\partial^2 z}{\partial^2 v}+2xy^{-3}\frac{\partial z}{\partial v}.$$

将结果代入方程$x^2 z_{xx}-y^2 z_{yy}=0$得 $2u\dfrac{\partial^2 z}{\partial u\partial v}-\dfrac{\partial z}{\partial v}=0$.

注 这题在变量关系上不容易厘清，计算z_{yy}时不少同学会漏掉项$2xy^{-3}\dfrac{\partial z}{\partial v}$. 而变换方程也是偏微分方程求解的基础！